

Current pathways for epidemiological research in amyotrophic lateral sclerosis.

Amyotrophic lateral sclerosis (ALS) is a rapidly progressive neurodegenerative disease. The current status of the epidemiology, challenges to its study, and novel study design options are discussed in this paper. We focus on recent results from large-scale population based prospective studies, case-control studies and population based registries, risk factors, and neuropathologic findings in chronic traumatic encephalomyelopathy. We identify areas of interest for future research, including time-trends in the incidence and prevalence of ALS; the meaning of lifetime risk; the phenotypic description of ALS; the definition of familial versus sporadic ALS, syndromic aspects of ALS; specific risk factors such as military service, life style factors such as smoking, the use of statins, and the presence of β -N-methylamino-L-alanine (BMAA), an excitotoxic amino acid derivative possibly produced by cyanobacteria found in almost every terrestrial and aquatic habitat; the emergence and disappearance of an endemic ALS in areas of the Pacific; and gene-environment interactions in the etiology of ALS. To move the epidemiology forward, we suggest using well-characterized cohorts of newly diagnosed ALS patients to identify risk and prognostic factors; storing biological material for future studies; building on the National ALS Registry as a resource of future studies; working in multidisciplinary consortia; and addressing the possible early life etiology of ALS.